ACYCLICITY AND SQUARE-ZERO KERNEL FOR THE ALMOST BLOWUP SQUARE OF AN EQUATION IN TWO GENERATORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EVJ (sources/stacks-flat/pinned-original-source.tex line 11782).

Allowed standard prerequisites (editorial).

- The almost blowup square of an equation in two generators. Same chapter, flat.tex, Example labelled example-two-generators (source line 11488), invoked in the statement and at proof line 11812. Supplies the square Z, X, X', E and the description X = Spec(A), X' = Proj(A[T_0,T_1]/(f_2T_0 - f_1T_1)) used throughout.
- Cohomology of the structure sheaf on a projective space over a ring. Cohomology of Schemes, Section labelled coherent-section-cohomology-projective-space (Stacks tag 01XS), invoked at proof lines 11804-11806. Supplies the vanishing and the global sections of the terms of the resolution in the universal case.
- Fibrewise criteria for the vanishing of H^1 and for the surjectivity of H^0 after base change. More on Morphisms, Lemmas labelled more-morphisms-lemma-check-h1-fibre-zero (Stacks tag 0E7F), more-morphisms-lemma-h1-fibre-zero (Stacks tag 0E7G) and more-morphisms-lemma-h1-fibre-zero-check-h0-kappa (Stacks tag 0E7I), invoked at proof lines 11814-11817. Supply the base change argument that transfers the universal computation to an arbitrary A; the kernel computation is done by hand inside the counted span.

Source reference. Original definition 0EVH: sources/stacks-flat/pinned-original-source.tex, source lines 11488–11488. Same-chapter pointer into the author environment example at source line 11488; invoked at proof line 11812 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 01XS: sources/stacks-chow/tags.txt, source lines 2531–2531. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 11806 of the counted proof.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

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Source reference. Original lemma 0E7F: sources/stacks-chow/tags.txt, source lines 17780–17780. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 11815 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E7G: sources/stacks-chow/tags.txt, source lines 17781–17781. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 11816 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E7I: sources/stacks-chow/tags.txt, source lines 17783–17783. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 11817 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let Z, X, X', E be an almost blow up square as in Example 0EVH. Then $H^p(X', \mathcal{O}_{X'}) = 0$ for $p > 0$ and $\Gamma(X, \mathcal{O}_X) \to \Gamma(X', \mathcal{O}_{X'})$ is a surjective map of rings whose kernel is an ideal of square zero.*

Proof. First assume that $A = \mathbf{Z}[f_1, f_2]$ is the polynomial ring. In this case our almost blow up square is the blowing up of $X = \mathrm{Spec}(A)$ in the closed subscheme Z and in fact $X' \subset \mathbf{P}^1_X$ is an effective Cartier divisor cut out by the global section $f_2T_0 - f_1T_1$ of $\mathcal{O}_{\mathbf{P}^1_X}(1)$. Thus we have a resolution

$$0 \to \mathcal{O}_{\mathbf{P}^1_X}(-1) \to \mathcal{O}_{\mathbf{P}^1_X} \to \mathcal{O}_{X'} \to 0$$

Using the description of the cohomology given in Cohomology of Schemes, Section 01XS it follows that in this case $\Gamma(X, \mathcal{O}_X) \to \Gamma(X', \mathcal{O}_{X'})$ is an isomorphism and $H^1(X', \mathcal{O}_{X'}) = 0$.

Next, we observe that any diagram as in Example 0EVH is the base change of the diagram in the previous paragraph by the ring map $\mathbf{Z}[f_1, f_2] \to A$. Hence by More on Morphisms, Lemmas 0E7F, 0E7G, and 0E7I we conclude that $H^1(X', \mathcal{O}_{X'})$ is zero in general and the surjectivity of the map $H^0(X, \mathcal{O}_X) \to H^0(X', \mathcal{O}_{X'})$ in general.

Next, in the general case, let us study the kernel. If $a \in A$ maps to zero, then looking on affine charts we see that

$$a = (f_1x - f_2)(a_0 + a_1x + \ldots + a_rx^r) \text{ in } A[x]$$

for some $r \geq 0$ and $a_0, \ldots, a_r \in A$ and similarly

$$a = (f_1 - f_2y)(b_0 + b_1y + \ldots + b_sy^s) \text{ in } A[y]$$

for some $s \geq 0$ and $b_0, \ldots, b_s \in A$. This means we have

$$a = f_2a_0,\ f_1a_0 = f_2a_1,\ \ldots,\ f_1a_r = 0,\ a = f_1b_0,\ f_2b_0 = f_1b_1,\ \ldots,\ f_2b_s = 0$$

If (a', r', a'_i, s', b'_j) is a second such system, then we have

$$aa' = f_1f_2a_0b'_0 = f_1f_2a_1b'_1 = f_1f_2a_2b'_2 = \ldots = 0$$

as desired. □